One-to-One and Many-to-Many Relationships

Lecture Notes · Game DB Junior

In this lesson

- The other two kinds of relationships: **one-to-one** and **many-to-many**
- Real world examples of each
- A quick look at all **four kinds** together

1. Two more kinds

Our photo sharing database only has **one-to-many** and **many-to-one** relationships. But there are two other important kinds: **many-to-many** and **one-to-one**. They are not in the database we are about to build. We learn them now so you know all four kinds exist. We will see examples of them later in the course.

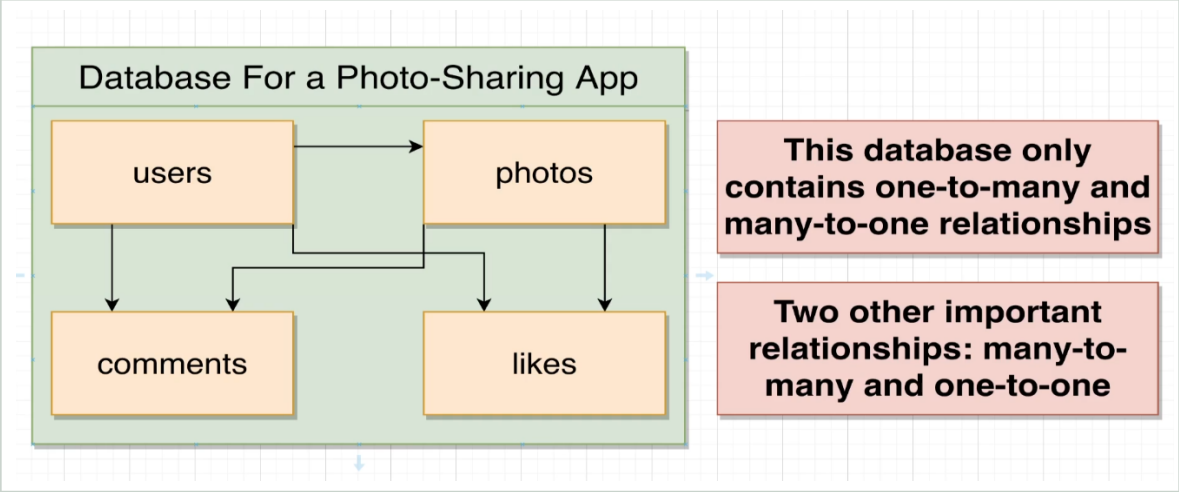


Figure 1 — Our database has only one-to-many and many-to-one relationships

Reminder: one-to-many / many-to-one

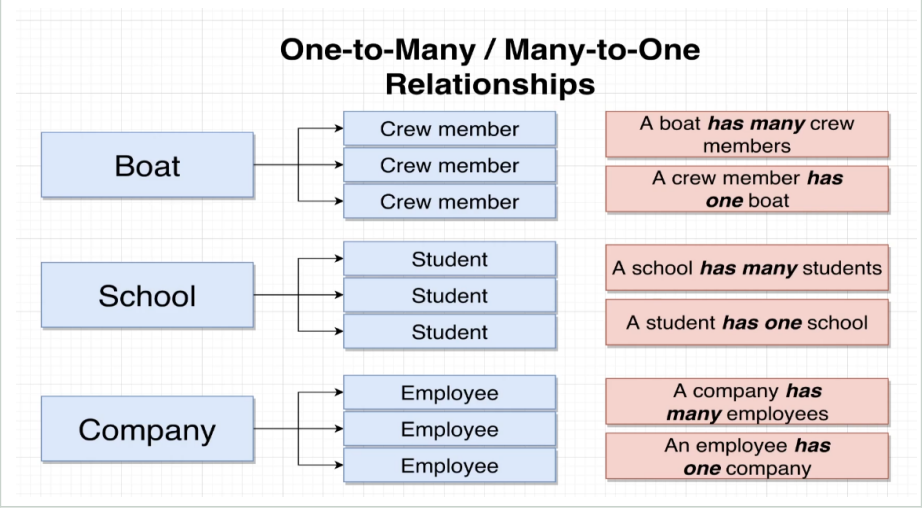


Figure 2 — Has many / has one (from the last lesson)

2. One-to-One

In a **one-to-one** relationship, **exactly one** record is related to **exactly one** other record, and the opposite is also true.

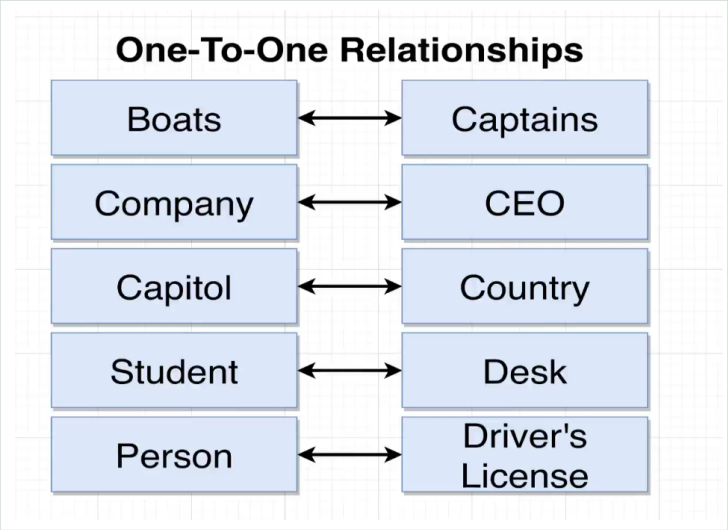


Figure 3 — One-to-one relationships

Pair	Why it is one-to-one
Boat ↔ Captain	A boat has one captain at a time. A captain is the captain of one boat at a time
Company ↔ CEO	A company has one CEO at a time. A CEO is the CEO of one company
Capital ↔ Country	No city is the capital of many countries at the same time. A capital has one country, and a country has one capital
Student ↔ Desk	A student sits at one desk, and a desk seats one student at a time
Person ↔ Driver's license	A person has one driver's license at a time. The license has the person's name on it, so it is only for that one person

There can be exceptions in the real world (for example, a company with many CEOs). For this discussion, we **generalize**.

3. Many-to-Many

In a **many-to-many** relationship, a record has many of the other records, and each of those also has many of the first. This is a bit trickier. We will see how to build it later in the course.

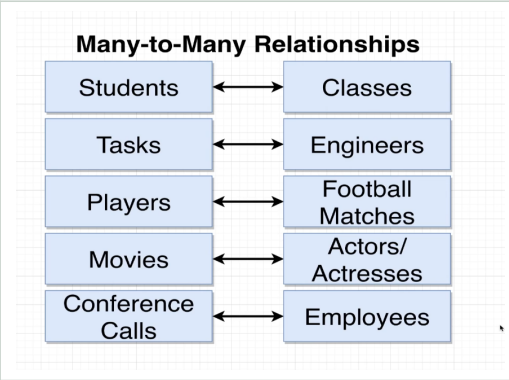


Figure 4 — Many-to-many relationships

Pair	Why it is many-to-many
Students ↔ Classes	A student has many classes, and a class has many students
Tasks ↔ Engineers	A task can have many engineers working on it, and an engineer can work on many tasks
Players ↔ Football matches	Many players play in one match, and a player plays in many matches
Movies ↔ Actors/Actresses	Most movies have many actors or actresses, and an actor or actress can star in many movies
Conference calls ↔ Employees	Over one week, a call has many employees, and an employee is in many calls

Time matters: in the conference call example, we look at a long time, like one week. At one single moment, an employee is only in one call. But over a week, an employee is in many calls.

4. Summary: the four kinds

Kind	Meaning	Example
One-to-many	One record has many others	A user has many photos
Many-to-one	Many records belong to one	A photo has one user
One-to-one	Exactly one record is related to exactly one other	A country has one capital
Many-to-many	Many records are related to many others	Students and classes

For the database we work on now, we only use **one-to-many** and **many-to-one**. We still do not know how to store the data that ties records together. **Next lesson:** how to set up a one-to-many or many-to-one relationship.

Sources: Lecture transcript and lecture images (Figures 1–4)